

Logic Emulation Machine (LEM) Business Case

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For low speed logic, such as an update rate less than 50 Kilo Hertz, FPGA implementations with 100 Mega Hertz clock rates allow two thousand logic operations between clocks. As memory is less resource consuming than logic (it takes sixty or more configuration bits to route a signal from a FPGA source to a single destination) it makes sense to devise an ISA (Instruction Set Architecture) to implement slow speed logic.

This application area is also handled by Programmed Logic Controllers (PLC). Here in the context of FPGA implementation, a logic processor for small numbers of signals has very low resource requirements: less than one hundred LUTs and half of a block RAM. This is less than 1% of a small FPGA chip. Prorated on low cost FPGAs it provides a LEM for three cents.

Benefits:

The FPGA tools provide timing analysis.

The LEM ISAs are completely deterministic (for known maximum instruction length).

FPGA simulation facilities provide off-line testing and verification.

Additional FPGA resources can be added for specialized needs.

The 4-bit data versions can do both binary and BCD addition/subtraction

Implementations:

LEM1_9	Bit serial, conditional branches and subroutine return stack
LEM1_9ptr	LEM1_9, ibid. with data memory pointers for increased generality
LEM4_9	Digit serial, conditional branches and subroutine return stack
LEM4_9ptr	LEM4_9 ibid. with data memory pointers for increased generality